LAB 7

Contrastive / Metric Learning on MagnaTagATune (MTAT)

Assignment objectives

1. Train a metric learning model on the MagnaTagATune (MTAT) dataset using contrastive learning (triplet loss and InfoNCE).
2. Evaluate the embedding quality:
   - Visualize embeddings in 2D for two non-overlapping tag groups (multi-label dataset).
   - Provide separate plots for each group.
3. Write a short report (in this notebook) describing decisions and observations.

Notebook steps (lecture alignment)

1. Train a CNN with triplet loss
2. Plot validation embeddings and clustering results
3. Train a CNN with InfoNCE loss
4. Plot validation embeddings and clustering results

Extras

- runnable on Modal, Colab, and local MacBook
- resource usage + environmental impact tracking
- basic economic cost analysis for training/inference

SETUP

```
In [1]: !pip -q install torchcodec
        !apt-get -y update -qq
        !apt-get -y install -qq ffmpeg
```

```
zsh:1: command not found: apt-get
zsh:1: command not found: apt-get
```

```
In [2]: import os, sys, subprocess, shutil, warnings, platform
        from pathlib import Path

        def running_on_colab():
            return "google.colab" in sys.modules

        def running_on_modal():
            return os.environ.get("MODAL_ENVIRONMENT", "") != ""

        def pip_install(pkgs):
            subprocess.check_call([sys.executable, "-m", "pip", "install", "-q"] + pkgs)

        # installs solo en Colab/Modal
        if running_on_colab() or running_on_modal():
            pip_install([
                "torch", "torchaudio",
                "tqdm", "scikit-learn", "matplotlib",
                "umap-learn",
                "psutil", "codecarbon",
                "pandas", "numpy",
                "soundfile"
            ])
        else:
            print("Local env -> skipping pip installs (use your venv).")

        # ffmpeg
        ff = shutil.which("ffmpeg")
        print("ffmpeg in PATH:", ff)

        if ff is None and (running_on_colab() or running_on_modal()):
            subprocess.check_call(["apt-get", "update", "-y"])
            subprocess.check_call(["apt-get", "install", "-y", "ffmpeg"])
            ff = shutil.which("ffmpeg")
            print("ffmpeg in PATH after install:", ff)

        print("python:", sys.version)
        print("platform:", platform.platform())
        print("running on colab:", running_on_colab())
        print("running on modal:", running_on_modal())
        if Path("/vol").is_dir():
            print("ls /vol:", os.listdir("/vol")[:30])

        warnings.filterwarnings("ignore", message="PySoundFile failed.*")
        warnings.filterwarnings("ignore", message=".*audioread.*")
        warnings.filterwarnings("ignore", message="n_fft=.* is too large.*")
```

```
Local env -> skipping pip installs (use your venv).
ffmpeg in PATH: /opt/homebrew/bin/ffmpeg
python: 3.11.14 (main, Oct  9 2025, 16:16:55) [Clang 17.0.0 (clang-1700.4.4.1)]
platform: macOS-26.1-arm64-arm-64bit
running on colab: False
running on modal: False
```

Setup and reproducibility

This section configures:

- device selection (cpu / cuda / mps)
- random seeds
- filesystem paths (Modal / Colab / local)
- lightweight logging utilities (timing + resource tracking)

```
In [3]: # setup device and paths
        import time, json, random, os, sys
        import numpy as np
        import torch

        SEED = 42
        random.seed(SEED)
        np.random.seed(SEED)
        torch.manual_seed(SEED)

        def running_on_colab():
            return "google.colab" in sys.modules

        def running_on_modal():
            return os.environ.get("MODAL_ENVIRONMENT", "") != ""

        def pick_device():
            if torch.cuda.is_available():
                return "cuda"
            if getattr(torch.backends, "mps", None) and torch.backends.mps.is_available():
                return "mps"
            return "cpu"

        DEVICE = pick_device()

        print("DEVICE:", DEVICE)
        print("torch default dtype:", torch.get_default_dtype())
        print(
            "mps available:",
            torch.backends.mps.is_available()
            if getattr(torch.backends, "mps", None)
            else None
        )

        # paths
        if running_on_modal():
            DATA_ROOT = "/mnt/mtat-datalabs"
            RUNS_ROOT = "/vol/runs" if os.path.isdir("/vol/runs") else "/mnt/runs"
        elif running_on_colab():
            DATA_ROOT = "/content/data/mtat"
            RUNS_ROOT = "/content/runs"
        else:
            DATA_ROOT = "./data/mtat"
            RUNS_ROOT = "./runs"

        os.makedirs(DATA_ROOT, exist_ok=True)
        os.makedirs(RUNS_ROOT, exist_ok=True)

        # environment summary
        print("environment summary")
        print("  running on colab:", running_on_colab())
        print("  running on modal:", running_on_modal())
        print("  torch.cuda.is_available:", torch.cuda.is_available())
        print("  mps available:", getattr(torch.backends, "mps", None) is not None and torch.backends.mps.is_available())
        print("  selected device:", DEVICE)
        print("  data root:", DATA_ROOT)
        print("  runs root:", RUNS_ROOT)
```

```
DEVICE: mps
torch default dtype: torch.float32
mps available: True
environment summary
  running on colab: False
  running on modal: False
  torch.cuda.is_available: False
  mps available: True
  selected device: mps
  data root: ./data/mtat
  runs root: ./runs
```

Resource and environmental impact tracking

Tracking of:

- wall-clock time
- CPU/RAM usage
- GPU memory (if available)
- estimated CO2 emissions using CodeCarbon (offline mode)

```
In [4]: # resource + carbon tracking
        import psutil
        from codecarbon import OfflineEmissionsTracker

        proc = psutil.Process(os.getpid())

        def get_resources_snapshot():
            snap = {
                "time": time.time(),
                "cpu_percent": psutil.cpu_percent(interval=None),
                "ram_mb": proc.memory_info().rss / (1024**2),
            }
            if torch.cuda.is_available():
                snap["gpu_mem_alloc_mb"] = torch.cuda.memory_allocated() / (1024**2)
                snap["gpu_mem_reserved_mb"] = torch.cuda.memory_reserved() / (1024**2)
            return snap

        class Timer:
            def __enter__(self):
                self.t0 = time.time()
                return self
            def __exit__(self, exc_type, exc, tb):
                self.dt = time.time() - self.t0

        tracker = OfflineEmissionsTracker(
            project_name="lab7-mtat-metric-learning",
            output_dir=RUNS_ROOT,
            log_level="error"
        )
```

```
[codecarbon INFO @ 16:05:51] offline tracker init
[codecarbon WARNING @ 16:05:51] Multiple instances of codecarbon are allowed to run at the same time.
```

Load MagnaTagATune (MTAT)

Mirdata is used to manage MTAT metadata and audio paths. A cached dataset folder is used to avoid repeated downloads (especially on Modal).

MTTAT loaded from local files (Lab 6 layout) to keep the notebook portable across Modal, Colab, and local runs.

Expected structure (auto-detected):

- datalabs/annotations_final.csv
- datalabs/clip_info_final.csv
- datalabs/audio/...

```
In [5]: from pathlib import Path
        import pandas as pd
        import numpy as np
        import os, sys

        def _has_mtat_layout(root: Path) -> bool:
            root = Path(root)
            return (
                (root / "annotations_final.csv").is_file()
                and (root / "clip_info_final.csv").is_file()
                and (root / "audio").is_dir()
            )

        def running_on_colab():
            return "google.colab" in sys.modules

        def running_on_modal():
            return os.environ.get("MODAL_ENVIRONMENT", "") != ""

        def find_upwards(start: Path, target: str, max_levels: int = 6):
            cur = start.resolve()
            for _ in range(max_levels):
                cand = cur / target
                if cand.exists():
                    return cand
                if cur.parent == cur:
                    break
                cur = cur.parent
            return None

        def find_in_roots(roots, filename="annotations_final.csv"):
            for root in roots:
                root = Path(root)
                if not root.exists():
                    continue
                for p in root.rglob(filename):
                    cand = p.parent
                    if _has_mtat_layout(cand):
                        return cand
            return None

        def _mount_drive_robust(mountpoint="/content/drive"):
            from pathlib import Path
            try:
                from google.colab import drive
            except Exception as e:
                raise RuntimeError(f"Not in Colab or drive module missing: {e}")

            mp = Path(mountpoint)
            already = mp.exists() and (mp / "MyDrive").exists()
            if already:
                print("Drive already mounted:", mountpoint)
                return

            drive.mount(mountpoint)
            if not (mp / "MyDrive").exists():
                drive.mount(mountpoint, force_remount=True)

        # ----------------------------
        # DATALABS_ROOT selection
        # ----------------------------
        DATALABS_ROOT = None

        # 0) Si estás en Mac/local y te quedó un env de Colab, lo ignoramos automáticamente
        override = os.environ.get("MTAT_DATALABS_ROOT", "").strip()
        if override and (not running_on_colab()) and (not running_on_modal()):
            if override.startswith("/content/"):
                override = ""  # no uses paths de Colab en Mac

        # 1) Optional override (solo si es válido)
        if override:
            cand = Path(override)
            if _has_mtat_layout(cand):
                DATALABS_ROOT = cand
            else:
                listing = sorted([p.name for p in cand.glob("*")])[:40] if cand.exists() else "path does not exist"
                print("MTAT_DATALABS_ROOT inválido (ignorado). Contenido:", listing)
                # NO reventamos acá: seguimos con autodetección
                DATALABS_ROOT = None

        # 2) Modal
        if DATALABS_ROOT is None and running_on_modal():
            candidates = [
                Path("/mnt/mtat-datalabs/datalabs"),
                Path("/mnt/mtat-datalabs"),
                Path("/vol/mtat/datalabs"),
                Path("/vol/mtat"),
                Path("/mnt/mtat/datalabs"),
                Path("/mnt/mtat"),
            ]
            for c in candidates:
                if _has_mtat_layout(c):
                    DATALABS_ROOT = c
                    break
            if DATALABS_ROOT is None:
                raise RuntimeError("MTAT datalabs not found on Modal (expected annotations_final.csv, clip_info_final.csv, audio/).")

        # 3) Colab: mount + search anywhere in MyDrive
        if DATALABS_ROOT is None and running_on_colab():
            _mount_drive_robust("/content/drive")
            search_roots = [
                Path("/content/drive/MyDrive"),
                Path("/content/drive/MyDrive/data"),
                Path("/content/drive/MyDrive/datalab5"),
                Path("/content/drive/MyDrive/data/datalab5"),
                Path("/content"),
            ]
            found = find_in_roots(search_roots, "annotations_final.csv")
            if found is None:
                raise RuntimeError(
                    "MTAT datalabs not found in Drive.\n"
                    "Busco annotations_final.csv debajo de MyDrive y no aparece.\n"
                    "Solución: asegurá que existe una carpeta 'datalabs/' que contenga:\n"
                    "  annotations_final.csv, clip_info_final.csv, audio/\n"
                )
            DATALABS_ROOT = found

        # 4) Local (Mac)
        if DATALABS_ROOT is None:
            DATALABS_ROOT = (
                find_upwards(Path.cwd(), "datalabs")
                or find_in_roots([Path.cwd(), Path.cwd().parent, Path.cwd().parent.parent], "annotations_final.csv")
                or Path("./datalabs")
            )
            if not _has_mtat_layout(DATALABS_ROOT):
                raise RuntimeError(
                    "Local MTAT datalabs not found.\n"
                    "Expected ./datalabs with annotations_final.csv, clip_info_final.csv and audio/.\n"
                )

        AUDIO_DIR = DATALABS_ROOT / "audio"
        ANN_CSV = DATALABS_ROOT / "annotations_final.csv"
        CLIP_CSV = DATALABS_ROOT / "clip_info_final.csv"

        print("datalabs root (USED):", DATALABS_ROOT)
        print("audio dir exists:", AUDIO_DIR.is_dir(), AUDIO_DIR)
        print("ann csv exists:", ANN_CSV.is_file(), ANN_CSV)
        print("clip csv exists:", CLIP_CSV.is_file(), CLIP_CSV)

        ann = pd.read_csv(ANN_CSV, sep="\t")
        clip = pd.read_csv(CLIP_CSV, sep="\t")

        if "clip_id" not in ann.columns or "clip_id" not in clip.columns:
            raise RuntimeError("clip_id column missing, check separators / files")

        if "mp3_path" not in ann.columns and "mp3_path" not in clip.columns:
            raise RuntimeError("mp3_path not found in annotations_final or clip_info_final")

        if "mp3_path" not in ann.columns:
            df = ann.merge(clip[["clip_id", "mp3_path"]], on="clip_id", how="left")
        else:
            df = ann.copy()

        df = df.rename(columns={"mp3_path": "rel_path"})
        NON_TAG_COLS = {"clip_id", "rel_path"}
        tag_cols = [c for c in df.columns if c not in NON_TAG_COLS]

        df = df.dropna(subset=["rel_path"]).reset_index(drop=True)
        df["rel_path"] = df["rel_path"].astype(str).str.strip().str.replace("\\", "/", regex=False)

        file_mask = df["rel_path"].apply(lambda rp: (AUDIO_DIR / rp).is_file())
        missing = int((~file_mask).sum())
        df = df[file_mask].reset_index(drop=True)

        rng = np.random.RandomState(SEED)
        perm = rng.permutation(len(df))
        n_val = int(len(df) * 0.1)
        val_idx = perm[:n_val]
        train_idx = perm[n_val:]
        train_df = df.iloc[train_idx].reset_index(drop=True)
        val_df = df.iloc[val_idx].reset_index(drop=True)

        print("train_df:", train_df.shape)
        print("val_df:", val_df.shape)
        print("num tags:", len(tag_cols))
```

```
datalabs root (USED): /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs
audio dir exists: True /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio
ann csv exists: True /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/annotations_final.csv
clip csv exists: True /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/clip_info_final.csv
train_df: (23277, 190)
val_df: (2586, 190)
num tags: 188
```

Contrastive dataset design (multi-label MTAT)

Two augmented views created of the same audio excerpt:

- view_1: random 3s chunk + augmentations
- view_2: another chunk near the first one (or same chunk with different augmentation)

Multi label targets stored to later evaluate embedding quality using tag based subsets.

```
In [6]: # audio utilities + augmentations

        import torchaudio
        import torch.nn as nn
        from torchaudio.transforms import MelSpectrogram

        TARGET_SR = 16000
        CLIP_SECONDS = 3
        N_SAMPLES = TARGET_SR * CLIP_SECONDS

        class FeatureExtractor(nn.Module):
            def __init__(self, n_mels=64, n_fft=1024, hop_length=256):
                super().__init__()
                self.mel = MelSpectrogram(
                    sample_rate=TARGET_SR,
                    n_fft=n_fft,
                    hop_length=hop_length,
                    n_mels=n_mels
                )

            def forward(self, x):
                x = self.mel(x)
                x = torch.log10(1 + 1000 * x)
                return x

        feature_extractor = FeatureExtractor().to(DEVICE).eval()

        def augment_wave(x):
            g = 0.9 + 0.2 * torch.rand(x.shape[0], device=x.device)
            x = x * g.unsqueeze(1)
            noise = 0.003 * torch.randn_like(x)
            return x + noise
```

```
In [7]: # ---- Mel extraction sanity check (device aware, no training dep) ----
        import torch, time

        print("torch.cuda.is_available:", torch.cuda.is_available())
        print("mps available:", getattr(torch.backends, "mps", None) is not None and torch.backends.mps.is_available())

        x = torch.randn(8, N_SAMPLES).to(DEVICE)

        t0 = time.time()
        with torch.no_grad():
            m = feature_extractor(x)
        dt = time.time() - t0

        print("mel shape:", tuple(m.shape))
        print("mel compute time:", round(dt, 3), "s")
```

```
torch.cuda.is_available: False
mps available: True
mel shape: (8, 64, 188)
mel compute time: 0.434 s
```

```
In [8]: import random
        import numpy as np
        import torch
        import torchaudio
        from torch.utils.data import Dataset
        from pathlib import Path
        import os, sys, subprocess, shutil

        def _ffmpeg_decode_to_float32_mono(path, target_sr=16000):
            ff = shutil.which("ffmpeg")
            if ff is None:
                raise RuntimeError("ffmpeg not found in PATH (Mac: brew install ffmpeg).")

            cmd = [
                ff, "-nostdin", "-v", "error",
                "-i", str(path),
                "-ac", "1",
                "-ar", str(int(target_sr)),
                "-f", "f32le",
                "pipe:1",
            ]

            try:
                raw = subprocess.check_output(cmd, stderr=subprocess.STDOUT)
            except subprocess.CalledProcessError as e:
                # archivo corrupto / inválido -> lo tratamos como "no decodificable"
                msg = e.output.decode("utf-8", errors="ignore")
                raise RuntimeError(f"ffmpeg decode failed for {path}\n{msg}")

            y = np.frombuffer(raw, dtype=np.float32)
            if y.size == 0:
                raise RuntimeError(f"ffmpeg decoded empty audio: {path}")
            return torch.from_numpy(y)

        print("torchaudio:", torchaudio.__version__)
        print("ffmpeg:", shutil.which("ffmpeg"))

        class MTATContrastiveDataset(Dataset):
            def __init__(self, df, audio_dir, tag_cols, max_retries=8):
                self.df = df.reset_index(drop=True)
                self.audio_dir = Path(audio_dir)
                self.tag_cols = list(tag_cols)
                self.max_retries = int(max_retries)

                self.tags = self.tag_cols
                self.tag_to_idx = {t: i for i, t in enumerate(self.tags)}
                self.id_col = "clip_id" if "clip_id" in self.df.columns else None

            def __len__(self):
                return len(self.df)

            def _resolve_path(self, rel_path):
                rp = str(rel_path).strip().replace("\\", "/")
                return self.audio_dir / rp

            def _load_audio(self, rel_path):
                p = self._resolve_path(rel_path)
                if not p.is_file():
                    raise FileNotFoundError(str(p))

                try:
                    wav, sr = torchaudio.load(str(p))
                    if wav.dim() == 2:
                        wav = wav.mean(dim=0)
                    if sr != TARGET_SR:
                        wav = torchaudio.functional.resample(wav, sr, TARGET_SR)
                    return wav.contiguous()
                except Exception:
                    y = _ffmpeg_decode_to_float32_mono(p, target_sr=TARGET_SR)
                    return y.contiguous()

            def _random_chunk(self, y):
                if y.numel() < N_SAMPLES:
                    return torch.nn.functional.pad(y, (0, N_SAMPLES - y.numel()))
                start = random.randint(0, y.numel() - N_SAMPLES)
                return y[start:start + N_SAMPLES]

            def __getitem__(self, idx):
                tries = 0
                cur_idx = int(idx)

                while True:
                    row = self.df.iloc[cur_idx]
                    try:
                        y = self._load_audio(row["rel_path"])
                        break
                    except Exception:
                        tries += 1
                        if tries >= self.max_retries:
                            raise
                        cur_idx = random.randint(0, len(self.df) - 1)

                c1 = self._random_chunk(y)
                c2 = self._random_chunk(y)

                v1 = augment_wave(c1.unsqueeze(0)).squeeze(0)
                v2 = augment_wave(c2.unsqueeze(0)).squeeze(0)

                tag_vec = torch.tensor(row[self.tag_cols].values.astype(np.float32))
                track_id = int(row[self.id_col]) if self.id_col is not None else int(cur_idx)

                return {"view_1": v1, "view_2": v2, "tags": tag_vec, "track_id": track_id}

        train_ds = MTATContrastiveDataset(train_df, AUDIO_DIR, tag_cols, max_retries=8)
        val_ds   = MTATContrastiveDataset(val_df,   AUDIO_DIR, tag_cols, max_retries=8)

        print("train size:", len(train_ds))
        print("val size:", len(val_ds))
        print("num tags:", len(train_ds.tags))

        _ = train_ds[0]
        print("audio decode check: OK")
```

```
torchaudio: 2.9.1
ffmpeg: /opt/homebrew/bin/ffmpeg
train size: 23277
val size: 2586
num tags: 188
audio decode check: OK
```

Embedding CNN

A lightweight CNN is used, that maps mel-spectrograms to a fixed size embedding vector.

Aiming to keep it small to run efficiently on a local MacBook.

```
In [9]: import torch.nn.functional as F

        class CNNEncoder(nn.Module):
            def __init__(self, dense_size=64, emb_dim=64):
                super().__init__()
                self.dense_size = dense_size
                self.conv1 = nn.Conv2d(1, dense_size//4, 3, padding=1)
                self.conv2 = nn.Conv2d(dense_size//4, dense_size//2, 3, padding=1)
                self.conv3 = nn.Conv2d(dense_size//2, dense_size, 3, padding=1)
                self.bn1 = nn.BatchNorm2d(dense_size//4)
                self.bn2 = nn.BatchNorm2d(dense_size//2)
                self.bn3 = nn.BatchNorm2d(dense_size)
                self.pool = nn.MaxPool2d(4, 4)
                self.fc1 = nn.Linear(dense_size, dense_size)
                self.fc2 = nn.Linear(dense_size, emb_dim, bias=False)

            def forward(self, x_mel):
                x = x_mel.unsqueeze(1)
                x = self.bn1(self.pool(F.relu(self.conv1(x))))
                x = self.bn2(self.pool(F.relu(self.conv2(x))))
                x = self.bn3(self.pool(F.relu(self.conv3(x))))
                x = x.mean(dim=[2,3])
                x = F.relu(self.fc1(x))
                z = self.fc2(x)
                z = F.normalize(z, dim=1)
                return z
```

Metric learning losses

We implement:

- triplet loss (hard negative mining within batch)
- InfoNCE (SimCLR-style symmetric loss)

```
In [10]: def triplet_loss_hard(z1, z2, margin=0.2):
             # z1 anchors, z2 positives
             dist = torch.cdist(z1, z2, p=2)
             pos = torch.diag(dist)
             dist = dist + torch.eye(dist.size(0), device=dist.device) * 1e9
             neg, _ = dist.min(dim=1)
             return F.relu(pos - neg + margin).mean()

         def info_nce(z1, z2, temp=0.4):
             sim12 = z1 @ z2.T
             sim21 = z2 @ z1.T
             labels = torch.arange(sim12.size(0), device=sim12.device)
             l12 = F.cross_entropy(sim12 / temp, labels)
             l21 = F.cross_entropy(sim21 / temp, labels)
             return 0.5 * (l12 + l21)
```

Training loop

- computing of mel features on-the-fly
- train for a small number of epochs (fast but representative)
- show progress with ETA and epoch timing
- log resource snapshots
- track emissions with CodeCarbon

```
In [11]: from tqdm import tqdm
         from torch.utils.data import DataLoader
         import numpy as np
         import time
         import torch
         import platform

         if torch.cuda.is_available():
             torch.backends.cudnn.benchmark = True
             try:
                 torch.backends.cuda.matmul.allow_tf32 = True
                 torch.backends.cudnn.allow_tf32 = True
             except Exception:
                 pass

         def pick_mel_device():
             if torch.cuda.is_available() and str(DEVICE) == "cuda":
                 return "cuda"
             return "cpu"

         MEL_DEVICE = pick_mel_device()
         print("DEVICE:", DEVICE, "| MEL_DEVICE:", MEL_DEVICE)

         # move feature extractor ONCE
         feature_extractor = feature_extractor.to(MEL_DEVICE).eval()

         # AMP only for CUDA (MPS AMP not used)
         USE_AMP = (str(DEVICE) == "cuda")
         scaler = torch.cuda.amp.GradScaler(enabled=USE_AMP)

         def compute_mel(x):
             # x: [B, N]
             # non_blocking helps on CUDA when pin_memory=True
             if str(MEL_DEVICE) == "cuda":
                 x = x.float().to(MEL_DEVICE, non_blocking=True)
             else:
                 x = x.float().to(MEL_DEVICE)

             with torch.no_grad():
                 m = feature_extractor(x).contiguous()

             # keep output on DEVICE for the model
             if str(DEVICE) == "cuda":
                 return m.to(DEVICE, non_blocking=True)
             return m.to(DEVICE)

         def _dataloader_kwargs():

             on_colab = running_on_colab()
             on_modal = running_on_modal()
             on_cuda  = (str(DEVICE) == "cuda")
             is_mac_local = (platform.system().lower() == "darwin") and (not on_colab) and (not on_modal)

             if is_mac_local:
                 return dict(num_workers=0, pin_memory=False)

             if on_colab:
                 num_workers = 2
                 # If random torchaudio decode Drive hangs num_workers = 0
             elif on_modal:
                 num_workers = 4
             else:
                 num_workers = 2

             kw = dict(
                 num_workers=num_workers,
                 pin_memory=on_cuda
             )

             if num_workers > 0:
                 kw["persistent_workers"] = True
                 kw["prefetch_factor"] = 2

             return kw

         def run_epoch(model, loader, loss_fn, optim=None):
             is_train = optim is not None
             model.train() if is_train else model.eval()

             losses = []
             t0 = time.time()

             pbar = tqdm(loader, leave=False)
             for batch in pbar:
                 # if CUDA + pin_memory, non_blocking helps
                 if str(DEVICE) == "cuda":
                     x1 = batch["view_1"].to(DEVICE, non_blocking=True).float()
                     x2 = batch["view_2"].to(DEVICE, non_blocking=True).float()
                 else:
                     x1 = batch["view_1"].to(DEVICE).float()
                     x2 = batch["view_2"].to(DEVICE).float()

                 m1 = compute_mel(x1)
                 m2 = compute_mel(x2)

                 if USE_AMP:
                     with torch.cuda.amp.autocast():
                         z1 = model(m1)
                         z2 = model(m2)
                         loss = loss_fn(z1, z2)
                 else:
                     z1 = model(m1)
                     z2 = model(m2)
                     loss = loss_fn(z1, z2)

                 if is_train:
                     optim.zero_grad(set_to_none=True)
                     if USE_AMP:
                         scaler.scale(loss).backward()
                         scaler.step(optim)
                         scaler.update()
                     else:
                         loss.backward()
                         optim.step()

                 losses.append(float(loss.detach().cpu().item()))
                 pbar.set_postfix({"loss": float(np.mean(losses))})

             dt = time.time() - t0
             return float(np.mean(losses)), dt

         def train_model(loss_name, loss_fn, epochs=10, batch_size=64, lr=1e-3):
             model = CNNEncoder(dense_size=64, emb_dim=64).to(DEVICE)
             print("model device:", next(model.parameters()).device)
             optim = torch.optim.Adam(model.parameters(), lr=lr)

             dl_kw = _dataloader_kwargs()
             print("DataLoader kwargs:", dl_kw)

             train_loader = DataLoader(train_ds, batch_size=batch_size, shuffle=True,  **dl_kw)
             val_loader   = DataLoader(val_ds,   batch_size=batch_size, shuffle=False, **dl_kw)

             history = {"loss_name": loss_name, "train_loss": [], "val_loss": [], "epoch_time_s": [], "resources": []}

             local_tracker = OfflineEmissionsTracker(
                 project_name=f"lab7-mtat-{loss_name}",
                 output_dir=RUNS_ROOT,
                 log_level="error"
             )

             local_tracker.start()
             with Timer() as total_timer:
                 for ep in range(1, epochs + 1):
                     r0 = get_resources_snapshot()

                     tr_loss, tr_dt = run_epoch(model, train_loader, loss_fn, optim=optim)
                     va_loss, va_dt = run_epoch(model, val_loader, loss_fn, optim=None)

                     r1 = get_resources_snapshot()

                     history["train_loss"].append(tr_loss)
                     history["val_loss"].append(va_loss)
                     history["epoch_time_s"].append(tr_dt + va_dt)
                     history["resources"].append({"epoch": ep, "before": r0, "after": r1})

                     print(f"epoch {ep}/{epochs} | train {tr_loss:.4f} | val {va_loss:.4f} | time {tr_dt + va_dt:.1f}s")

             emissions = local_tracker.stop()
```

```
    history["total_time_s"] = total_timer.dt
    history["emissions_kgco2"] = emissions

    return model, history
```

```
DEVICE: mps | MEL_DEVICE: cpu
/var/folders/_1/5vl1nfxd15nd63gj1hbqvwq00000gn/T/ipykernel_31562/136063290.py:29: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use `torch.amp.GradScaler('cuda', arg
s...)` instead.
  scaler = torch.cuda.amp.GradScaler(enabled=USE_AMP)
```

Step 1 — Train with triplet loss

In [12]:

```python
triplet_model, triplet_hist = train_model(
    "triplet_hard",
    lambda a,b: triplet_loss_hard(a,b,margin=0.15),
    epochs=10,
    batch_size=64,
    lr=1e-3
)
```

```
model device: mps:0
DataLoader kwargs: {'num_workers': 0, 'pin_memory': False}

epoch 1/10 | train 0.1184 | val 0.0945 | time 254.3s

epoch 2/10 | train 0.0859 | val 0.0798 | time 248.9s

epoch 3/10 | train 0.0768 | val 0.0768 | time 248.9s

epoch 4/10 | train 0.0727 | val 0.0713 | time 249.0s

epoch 5/10 | train 0.0671 | val 0.0707 | time 249.4s

epoch 6/10 | train 0.0639 | val 0.0661 | time 276.4s

epoch 7/10 | train 0.0631 | val 0.0667 | time 268.9s

epoch 8/10 | train 0.0623 | val 0.0629 | time 262.5s

epoch 9/10 | train 0.0593 | val 0.0580 | time 262.1s

epoch 10/10 | train 0.0570 | val 0.0571 | time 265.8s
```

Step 2 — Embedding extraction and 2D visualization

Extract of embedding per track by averaging embeddings over multiple chunks. Then:

- run UMAP to 2D
- run k-means for a coarse clustering view
- create separate plots for two non-overlapping tag groups

In [13]:

```python
import umap
from sklearn.cluster import KMeans

def extract_track_embeddings(model, dataset, chunks_per_track=3):
    model.eval()
    feature_extractor.eval()

    Z, Y, T = [], [], []

    with torch.no_grad():
        for i in tqdm(range(len(dataset)), desc="embedding", leave=False):
            item = dataset[i]
            tid = item["track_id"]
            tags = item["tags"].cpu().numpy()

            zs = []
            for _ in range(chunks_per_track):
                x = item["view_1"].unsqueeze(0).to(DEVICE).float()
                m = compute_mel(x)
                z = model(m)
                zs.append(z.squeeze(0).cpu().numpy())

            Z.append(np.mean(np.stack(zs), axis=0))
            Y.append(tags)
            T.append(tid)

    return np.stack(Z), np.stack(Y), np.array(T)

Z_val_triplet, Y_val, T_val = extract_track_embeddings(
    triplet_model,
    val_ds,
    chunks_per_track=2
)
print("embeddings:", Z_val_triplet.shape, "labels:", Y_val.shape)
```

```
/Users/meco/MBP Files/Gits/upfs-virtualenv/lib/python3.11/site-packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedo
cs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
embeddings: (2586, 64) labels: (2586, 188)
```

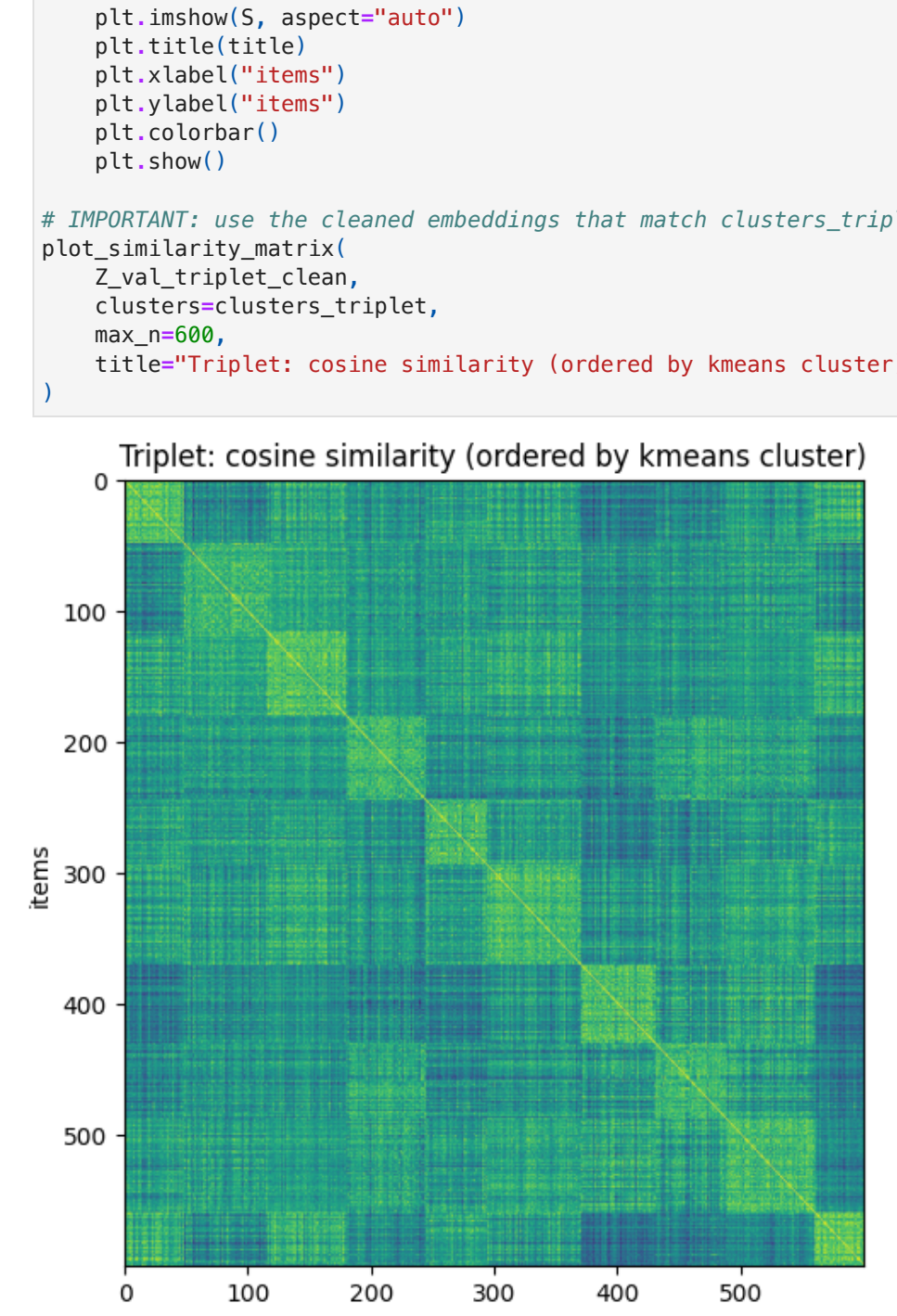
In [14]:

```python
import numpy as np
import umap
from sklearn.cluster import KMeans


Z = Z_val_triplet.astype(np.float32)
Y = Y_val
T = T_val

bad = ~np.isfinite(Z).all(axis=1)
num_bad = int(bad.sum())

print("total embeddings:", len(Z))
print("invalid embeddings (nan/inf):", num_bad)

if num_bad > 0:
    print("removing invalid embeddings before UMAP/KMeans")
    Z = Z[~bad]
    Y = Y[~bad]
    T = T[~bad]

Z_val_triplet_clean = Z
Y_val_clean = Y
T_val_clean = T

reducer = umap.UMAP(
    n_components=2,
    random_state=SEED,
    n_neighbors=15,
    min_dist=0.1
)

Z2_triplet = reducer.fit_transform(Z_val_triplet_clean)

kmeans = KMeans(
    n_clusters=10,
    random_state=SEED,
    n_init="auto"
)

clusters_triplet = kmeans.fit_predict(Z_val_triplet_clean)

print("umap:", Z2_triplet.shape)
print("clusters:", clusters_triplet.shape)
```

```
/Users/meco/MBP Files/Gits/upfs-virtualenv/lib/python3.11/site-packages/umap/umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting random_state. Use no seed for parallelism.
  warn(
total embeddings: 2586
invalid embeddings (nan/inf): 0
umap: (2586, 2)
clusters: (2586,)
```

In [15]:

```python
import numpy as np
import matplotlib.pyplot as plt

def plot_similarity_matrix(Z, clusters=None, max_n=600, title=""):
    Z = Z.astype(np.float32)
    Z = Z / (np.linalg.norm(Z, axis=1, keepdims=True) + 1e-9)

    n = min(len(Z), int(max_n))
    idx = np.arange(n)

    if clusters is not None:
        if len(clusters) != len(Z):
            raise RuntimeError(f"clusters and Z misaligned: clusters={len(clusters)} vs Z={len(Z)}")
        order = np.argsort(clusters[:n])
        idx = idx[order]

    S = Z[idx] @ Z[idx].T

    plt.figure(figsize=(7,6))
    plt.imshow(S, aspect="auto")
    plt.title(title)
    plt.xlabel("items")
    plt.ylabel("items")
    plt.colorbar()
    plt.show()

# IMPORTANT: use the cleaned embeddings that match clusters_triplet
plot_similarity_matrix(
    Z_val_triplet_clean,
    clusters=clusters_triplet,
    max_n=600,
    title="Triplet: cosine similarity (ordered by kmeans cluster)"
)
```



In [16]:

```python
import os, sys, subprocess, shutil
import numpy as np
import soundfile as sf
from pathlib import Path

def can_inline_audio():
    try:
        from IPython import get_ipython
        return get_ipython() is not None
    except Exception:
        return False

def _ffmpeg_decode_to_float32_mono(path, target_sr=16000):
    ff = shutil.which("ffmpeg")
    if ff is None:
        raise RuntimeError("ffmpeg not found in PATH (Mac: brew install ffmpeg).")

    cmd = [
        ff, "-nostdin", "-v", "error",
        "-i", str(path),
        "-ac", "1",
        "-ar", str(int(target_sr)),
        "-f", "f32le",
        "pipe:1",
    ]
    raw = subprocess.check_output(cmd)
    y = np.frombuffer(raw, dtype=np.float32)
    if y.size == 0:
        raise RuntimeError(f"ffmpeg decoded empty audio: {path}")
    return y

def _safe_audio_path(dataset, idx):
    rel = dataset.df.iloc[int(idx)]["rel_path"]
    return str(dataset.audio_dir / str(rel))

def _load_audio_for_play(path, sr=16000, seconds=6):
    y = _ffmpeg_decode_to_float32_mono(path, target_sr=sr)
    y = y[: int(sr * seconds)]
    return y

def knn_indices(Z, query_idx, k=6):
    Z = Z.astype(np.float32)
    Z = Z / (np.linalg.norm(Z, axis=1, keepdims=True) + 1e-9)
    sims = Z @ Z[int(query_idx)]
    sims[int(query_idx)] = -np.inf
    nn = np.argsort(-sims)[:k]
    return nn, sims[nn]

def show_query_and_neighbors(dataset, Z, query_idx, k=6, seconds=6, title="", out_subdir="qual_retrieval_triplet"):
    out_dir = Path(RUNS_ROOT) / out_subdir
    out_dir.mkdir(parents=True, exist_ok=True)

    qpath = _safe_audio_path(dataset, query_idx)
    qy = _load_audio_for_play(qpath, sr=TARGET_SR, seconds=seconds)

    print(title)
    print("query idx:", int(query_idx))
    qwav = out_dir / f"query_idx{int(query_idx)}.wav"
    sf.write(str(qwav), qy, TARGET_SR)
    print("query wav:", str(qwav))

    nn, sims = knn_indices(Z, query_idx, k=k)

    if can_inline_audio():
        from IPython.display import Audio, display, HTML
        display(HTML("<b>Query audio:</b>"))
        display(Audio(qy, rate=TARGET_SR))

    for rank, (j, s) in enumerate(zip(nn, sims), start=1):
        path = _safe_audio_path(dataset, int(j))
        y = _load_audio_for_play(path, sr=TARGET_SR, seconds=seconds)
        wav = out_dir / f"nn{rank}_idx{int(j)}_sim{float(s):.3f}.wav"
        sf.write(str(wav), y, TARGET_SR)
        print("nn", rank, "| idx:", int(j), "| sim:", float(s), "| wav:", str(wav))

        if can_inline_audio():
            from IPython.display import Audio, display, HTML
            display(HTML(f"<b>Neighbor {rank} (sim={float(s):.3f}):</b>"))
            display(Audio(y, rate=TARGET_SR))

def pick_query_with_tag(Y, dataset, tag_name):
    if tag_name not in dataset.tag_to_idx:
        return None
    ti = dataset.tag_to_idx[tag_name]
    idxs = np.where(Y[:, ti] == 1)[0]
    if len(idxs) == 0:
        return None
    return int(np.random.choice(idxs))

np.random.seed(SEED)
qidx = np.random.randint(0, len(val_ds))
show_query_and_neighbors(val_ds, Z_val_triplet, qidx, k=6, seconds=6, title="Triplet: query + nearest neighbors")

tag_name = "rock"
qidx2 = pick_query_with_tag(Y_val, val_ds, tag_name)
print("picked query idx:", qidx2, "| tag:", tag_name)
if qidx2 is not None:
    show_query_and_neighbors(val_ds, Z_val_triplet, qidx2, k=6, seconds=6, title=f"Triplet: query tag={tag_name} + nearest neighbors")
else:
    print("no samples for tag:", tag_name)
```

```
Triplet: query + nearest neighbors
query idx: 860
query wav: runs/qual_retrieval_triplet/query_idx860.wav
```

Query audio:

▶ 0:00 0:06

```
nn 1 | idx: 1335 | sim: 0.9154337644577026 | wav: runs/qual_retrieval_triplet/nn1_idx1335_sim0.915.wav
```

Neighbor 1 (sim=0.915):

▶ 0:00 0:06

```
nn 2 | idx: 1090 | sim: 0.9053992033004761 | wav: runs/qual_retrieval_triplet/nn2_idx1090_sim0.905.wav
```

Neighbor 2 (sim=0.905):

▶ 0:00 0:06

```
nn 3 | idx: 1723 | sim: 0.8969926834106445 | wav: runs/qual_retrieval_triplet/nn3_idx1723_sim0.897.wav
```

Neighbor 3 (sim=0.897):

▶ 0:00 0:06

```
nn 4 | idx: 793 | sim: 0.879464864730835 | wav: runs/qual_retrieval_triplet/nn4_idx793_sim0.879.wav
```

Neighbor 4 (sim=0.879):

▶ 0:00 0:06

```
nn 5 | idx: 2383 | sim: 0.8762547969818115 | wav: runs/qual_retrieval_triplet/nn5_idx2383_sim0.876.wav
```

Neighbor 5 (sim=0.876):

▶ 0:00 0:06

```
nn 6 | idx: 419 | sim: 0.8739794492721558 | wav: runs/qual_retrieval_triplet/nn6_idx419_sim0.874.wav
```

Neighbor 6 (sim=0.874):

▶ 0:00 0:06

```
picked query idx: 117 | tag: rock
Triplet: query tag=rock + nearest neighbors
query idx: 117
query wav: runs/qual_retrieval_triplet/query_idx117.wav
```

Query audio:

▶ 0:00 0:06

```
nn 1 | idx: 1533 | sim: 0.9475681781768799 | wav: runs/qual_retrieval_triplet/nn1_idx1533_sim0.948.wav
```

Neighbor 1 (sim=0.948):

▶ 0:00 0:06

```
nn 2 | idx: 500 | sim: 0.9406097531318665 | wav: runs/qual_retrieval_triplet/nn2_idx500_sim0.941.wav
```

Neighbor 2 (sim=0.941):

▶ 0:00 0:06

```
nn 3 | idx: 2370 | sim: 0.9362567067146301 | wav: runs/qual_retrieval_triplet/nn3_idx2370_sim0.936.wav
```

Neighbor 3 (sim=0.936):

▶ 0:00 0:06

```
nn 4 | idx: 145 | sim: 0.9296431541442871 | wav: runs/qual_retrieval_triplet/nn4_idx145_sim0.930.wav
```

Neighbor 4 (sim=0.930):

▶ 0:00 0:06

```
nn 5 | idx: 1973 | sim: 0.9245335459709167 | wav: runs/qual_retrieval_triplet/nn5_idx1973_sim0.925.wav
```

Neighbor 5 (sim=0.925):

▶ 0:00 -0:06

```
nn 6 | idx: 991 | sim: 0.9215233325958252 | wav: runs/qual_retrieval_triplet/nn6_idx991_sim0.922.wav
```

Neighbor 6 (sim=0.922):

▶ 0:00 -0:06

Tag groups (non-overlapping)

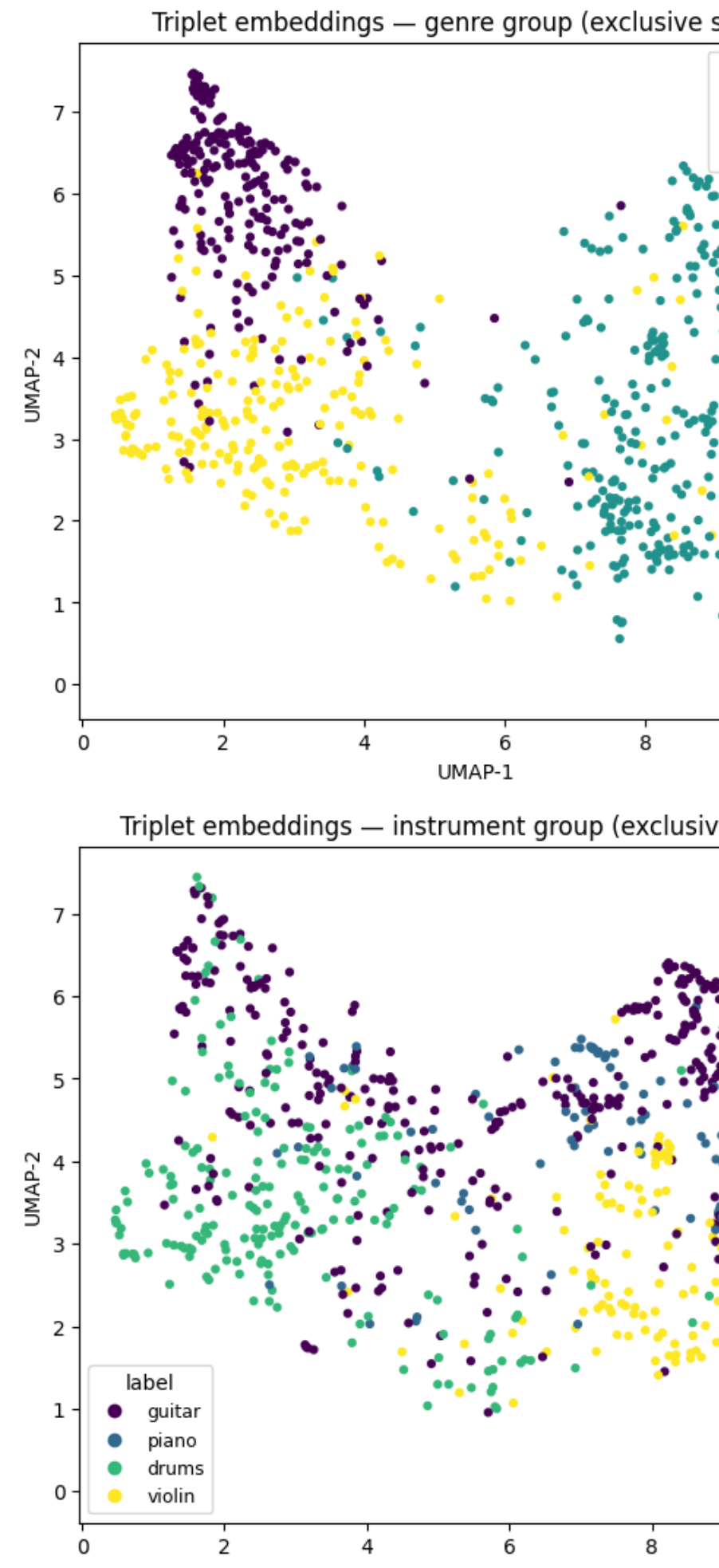
MTAT is multi-label, so we construct **exclusive subsets** for visualization:

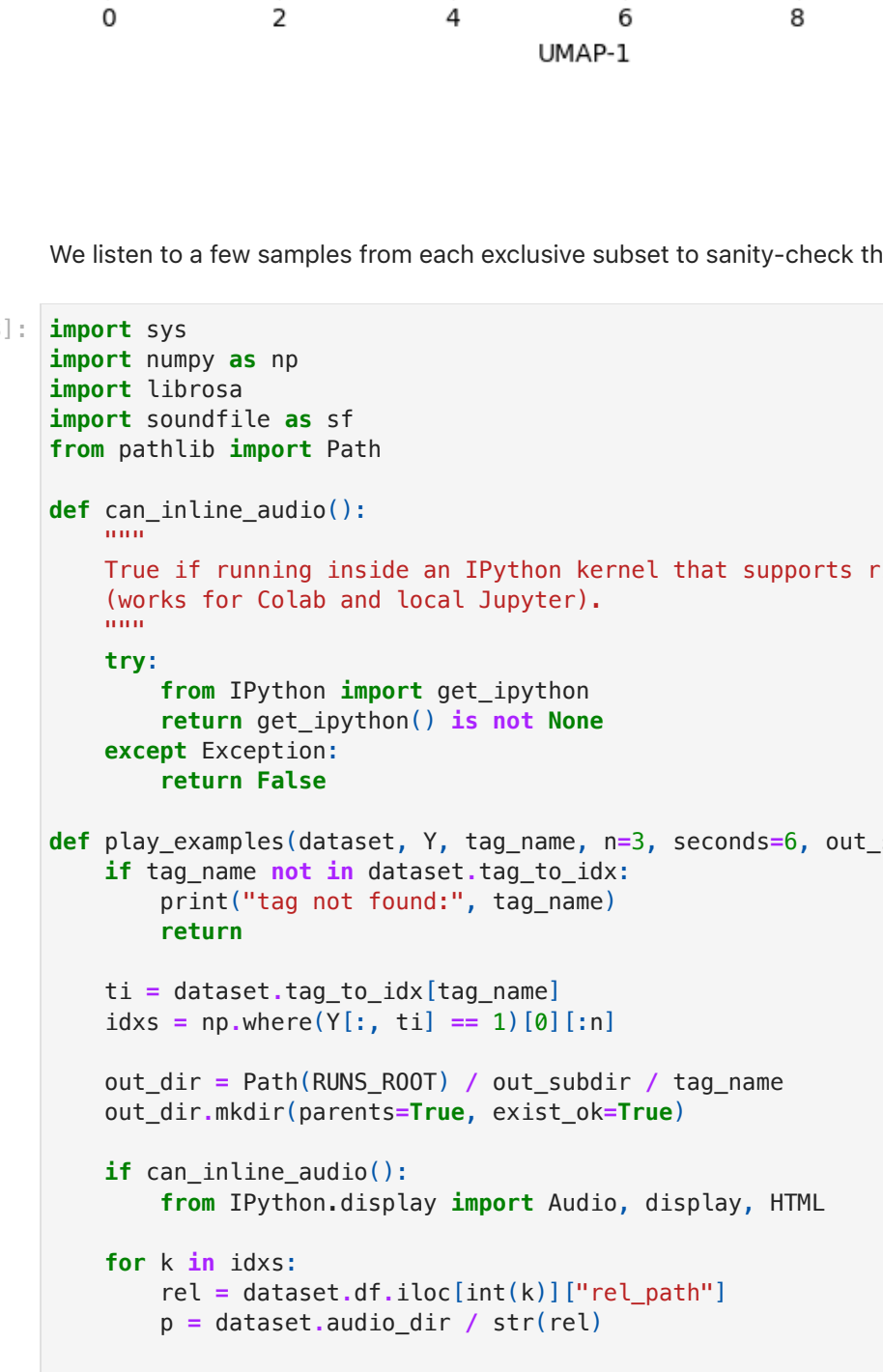
- Group A (genres): pick a few genre tags and keep only samples where exactly one of them is active.
- Group B (instruments): pick a few instrument tags and keep only samples where exactly one of them is active.

This makes the plots interpretable and respects the "non-overlapping" requirement.

In [17]:

```python
import matplotlib.pyplot as plt
import numpy as np

tags_list = train_ds.tags
tag_to_idx = train_ds.tag_to_idx

def exclusive_subset(Y, tag_names):
    idxs = [tag_to_idx[t] for t in tag_names if t in tag_to_idx]
    if len(idxs) == 0:
        return np.array([], dtype=bool), idxs
    sub = Y[:, idxs]
    one_hot = (sub.sum(axis=1) == 1)
    return one_hot, idxs

def plot_umap_for_group(Z2, Y_aligned, tag_names, title):
    if len(Z2) != len(Y_aligned):
        raise RuntimeError(
            f"Z2 and Y are misaligned: Z2={len(Z2)} vs Y={len(Y_aligned)}"
        )

    mask, idxs = exclusive_subset(Y_aligned, tag_names)
    if mask.size == 0 or mask.sum() == 0:
        print("no samples for:", title)
        return

    subZ2 = Z2[mask]
    subY = Y_aligned[mask][:, idxs]
    label_idx = subY.argmax(axis=1)
    label_names = [tag_names[i] for i in range(len(idxs))]

    plt.figure(figsize=(7,6))
    sc = plt.scatter(subZ2[:,0], subZ2[:,1], c=label_idx, s=12)
    plt.title(title)
    plt.xlabel("UMAP-1")
    plt.ylabel("UMAP-2")

    handles, _ = sc.legend_elements()
    plt.legend(handles, label_names, title="label", loc="best", fontsize=9)
    plt.show()

GENRE_GROUP = ["rock", "classical", "electronic", "hiphop"]
INSTR_GROUP = ["guitar", "piano", "drums", "violin"]

plot_umap_for_group(
    Z2_triplet,
    Y_val_clean,
    GENRE_GROUP,
    "Triplet embeddings — genre group (exclusive subset)"
)

plot_umap_for_group(
    Z2_triplet,
    Y_val_clean,
    INSTR_GROUP,
    "Triplet embeddings — instrument group (exclusive subset)"
)
```





Qualitative check: audio examples

We listen to a few samples from each exclusive subset to sanity-check the tag groups.

In [18]:

```python
import sys
import numpy as np
import librosa
import soundfile as sf
from pathlib import Path

def can_inline_audio():
    """
    True if running inside an IPython kernel that supports rich display
    (works for Colab and Jupyter).
    """
    try:
        from IPython import get_ipython
        return get_ipython() is not None
    except Exception:
        return False

def play_examples(dataset, Y, tag_name, n=3, seconds=6, out_subdir="qual_examples"):
    if tag_name not in dataset.tag_to_idx:
        print("tag not found:", tag_name)
        return

    ti = dataset.tag_to_idx[tag_name]
    idxs = np.where(Y[:, ti] == 1)[0][:n]

    out_dir = Path(RUNS_ROOT) / out_subdir / tag_name
    out_dir.mkdir(parents=True, exist_ok=True)

    if can_inline_audio():
        from IPython.display import Audio, display, HTML

    for k in idxs:
        rel = dataset.df.iloc[int(k)]["rel_path"]
        p = dataset.audio_dir / str(rel)

        y, _ = librosa.load(str(p), sr=TARGET_SR, mono=True, duration=seconds)
        wav = out_dir / f"idx{int(k)}.wav"
        sf.write(str(wav), y, TARGET_SR)

        if "clip_id" in dataset.df.columns:
            print("clip_id:", int(dataset.df.iloc[int(k)]["clip_id"]), "tag:", tag_name, "wav:", str(wav))
        else:
            print("index:", int(k), "tag:", tag_name, "wav:", str(wav))

        if can_inline_audio():
            display(HTML(f"<b>{tag_name} example idx={int(k)}</b>"))
            display(Audio(y, rate=TARGET_SR))
```

Step 3 — Train with InfoNCE loss

In [19]:

```python
infonce_model, infonce_hist = train_model(
    "infonce",
    lambda a,b: info_nce(a,b,temp=0.4),
    epochs=8,
    batch_size=64,
    lr=1e-3
)
```

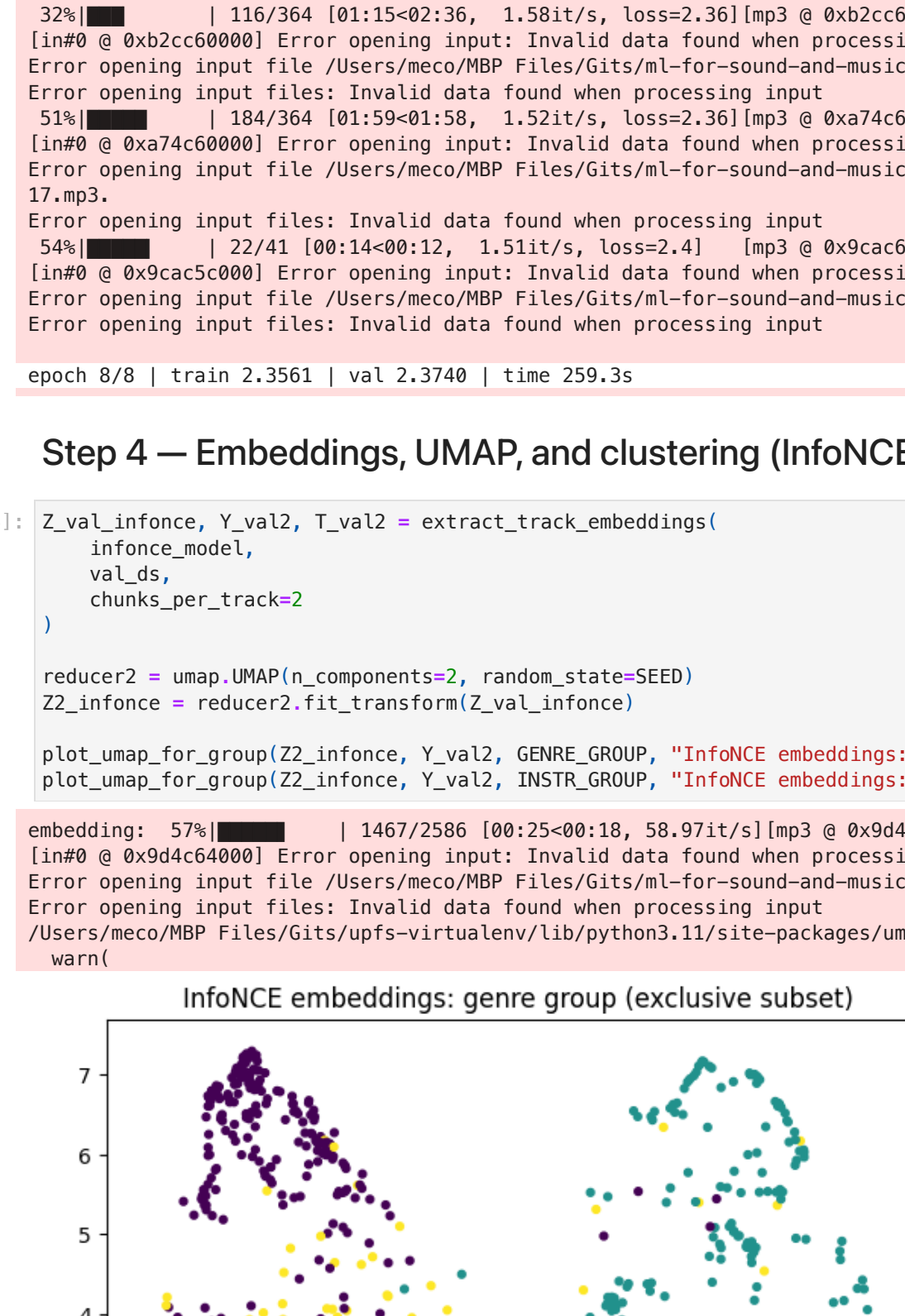
```
model device: mps:0
DataLoader kwargs: {'num_workers': 0, 'pin_memory': False}
 39%|███▉      | 141/364 [01:29<02:17,  1.63it/s, loss=2.76][mp3 @ 0x74f060000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x74ec1c000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/9/american_baroque-dances_and_suites_of_rameau_and_couperin-25-le_petit_rien_xiveme_ordre_couperin-88-1
17.mp3.
Error opening input files: Invalid data found when processing input
 40%|███▉      | 145/364 [01:32<02:16,  1.60it/s, loss=2.76][mp3 @ 0x7acc64000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x7acc60000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/8/jacob_heringman-josquin_des_prez_lute_settings-19-gintzler__pater_noster-204-233.mp3.
Error opening input files: Invalid data found when processing input
 84%|████████▍ | 306/364 [03:11<00:35,  1.63it/s, loss=2.65][mp3 @ 0xc80c48000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xc80c44000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/e/solace-balance-04-djinn-146-175.mp3.
Error opening input files: Invalid data found when processing input
 54%|█████▎    | 22/41 [00:13<00:11,  1.65it/s, loss=2.56]  [mp3 @ 0x8c2c64000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x8c2c60000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/6/norine_braun-now_and_zen-08-gently-117-146.mp3.
Error opening input files: Invalid data found when processing input

epoch 1/8 | train 2.6279 | val 2.5200 | time 252.0s
 35%|███▌      | 129/364 [01:20<02:25,  1.61it/s, loss=2.47][mp3 @ 0x85f03c000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x85ec3c000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/9/american_baroque-dances_and_suites_of_rameau_and_couperin-25-le_petit_rien_xiveme_ordre_couperin-88-1
17.mp3.
Error opening input files: Invalid data found when processing input
 48%|████▊     | 176/364 [01:51<02:05,  1.50it/s, loss=2.47][mp3 @ 0xb6cc14000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xb6d00c000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/8/jacob_heringman-josquin_des_prez_lute_settings-19-gintzler__pater_noster-204-233.mp3.
Error opening input files: Invalid data found when processing input
 61%|██████    | 222/364 [02:21<01:36,  1.46it/s, loss=2.46][mp3 @ 0xb86c60000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xb8700c000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/e/solace-balance-04-djinn-146-175.mp3.
Error opening input files: Invalid data found when processing input
 54%|█████▎    | 22/41 [00:13<00:12,  1.58it/s, loss=2.49]  [mp3 @ 0x858c68000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x858c64000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/6/norine_braun-now_and_zen-08-gently-117-146.mp3.
Error opening input files: Invalid data found when processing input

epoch 2/8 | train 2.4549 | val 2.4551 | time 263.7s
 32%|███▏      | 118/364 [01:16<02:39,  1.54it/s, loss=2.43][mp3 @ 0xc3cc48000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xc3cc44000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/8/jacob_heringman-josquin_des_prez_lute_settings-19-gintzler__pater_noster-204-233.mp3.
Error opening input files: Invalid data found when processing input
 48%|████▊     | 174/364 [01:53<02:05,  1.52it/s, loss=2.43][mp3 @ 0x7d8c64000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x7d9014000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/9/american_baroque-dances_and_suites_of_rameau_and_couperin-25-le_petit_rien_xiveme_ordre_couperin-88-1
17.mp3.
Error opening input files: Invalid data found when processing input
 82%|████████▏ | 300/364 [03:17<00:42,  1.51it/s, loss=2.42][mp3 @ 0x8f8c48000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x8f8c44000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/e/solace-balance-04-djinn-146-175.mp3.
Error opening input files: Invalid data found when processing input
 54%|█████▎    | 22/41 [00:14<00:12,  1.57it/s, loss=2.48]  [mp3 @ 0x9e484c000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x9e4848000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/6/norine_braun-now_and_zen-08-gently-117-146.mp3.
Error opening input files: Invalid data found when processing input

epoch 3/8 | train 2.4177 | val 2.4634 | time 267.1s
  2%|▏         | 6/364 [00:04<04:06,  1.45it/s, loss=2.34][mp3 @ 0x71ec44000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x71ec40000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/9/american_baroque-dances_and_suites_of_rameau_and_couperin-25-le_petit_rien_xiveme_ordre_couperin-88-1
17.mp3.
Error opening input files: Invalid data found when processing input
 29%|██▊       | 104/364 [01:10<02:54,  1.49it/s, loss=2.4][mp3 @ 0x766c48000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x766c44000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/8/jacob_heringman-josquin_des_prez_lute_settings-19-gintzler__pater_noster-204-233.mp3.
Error opening input files: Invalid data found when processing input
 49%|████▊     | 177/364 [01:59<02:04,  1.50it/s, loss=2.4][mp3 @ 0x762808000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x762804000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/e/solace-balance-04-djinn-146-175.mp3.
Error opening input files: Invalid data found when processing input
 54%|█████▎    | 22/41 [00:14<00:12,  1.54it/s, loss=2.43] [mp3 @ 0xbf6814000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xbf7450000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/6/norine_braun-now_and_zen-08-gently-117-146.mp3.
Error opening input files: Invalid data found when processing input

epoch 4/8 | train 2.3969 | val 2.4076 | time 271.3s
 22%|██▏       | 79/364 [00:52<03:09,  1.50it/s, loss=2.38][mp3 @ 0x95e804000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x95f404000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/e/solace-balance-04-djinn-146-175.mp3.
Error opening input files: Invalid data found when processing input
 70%|███████   | 255/364 [02:52<01:15,  1.44it/s, loss=2.39][mp3 @ 0xadac34000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xadb030000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/8/jacob_heringman-josquin_des_prez_lute_settings-19-gintzler__pater_noster-204-233.mp3.
Error opening input files: Invalid data found when processing input
 72%|███████▏  | 261/364 [02:57<01:12,  1.41it/s, loss=2.39][mp3 @ 0x9d8c44000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x9d9034000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/9/american_baroque-dances_and_suites_of_rameau_and_couperin-25-le_petit_rien_xiveme_ordre_couperin-88-1
17.mp3.
Error opening input files: Invalid data found when processing input
 54%|█████▎    | 22/41 [00:15<00:13,  1.45it/s, loss=2.45]  [mp3 @ 0xcb8c68000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xcb8c64000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/6/norine_braun-now_and_zen-08-gently-117-146.mp3.
Error opening input files: Invalid data found when processing input

epoch 5/8 | train 2.3828 | val 2.4117 | time 275.3s
 55%|█████▌    | 201/364 [02:18<01:52,  1.45it/s, loss=2.37][mp3 @ 0x8d2c44000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x8d3034000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/e/solace-balance-04-djinn-146-175.mp3.
Error opening input files: Invalid data found when processing input
 57%|█████▋    | 207/364 [02:22<01:48,  1.44it/s, loss=2.37][mp3 @ 0xb10c64000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xb10c60000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/8/jacob_heringman-josquin_des_prez_lute_settings-19-gintzler__pater_noster-204-233.mp3.
Error opening input files: Invalid data found when processing input
 97%|█████████▋| 352/364 [03:58<00:07,  1.51it/s, loss=2.37][mp3 @ 0x7b2c18000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x7b2c14000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/9/american_baroque-dances_and_suites_of_rameau_and_couperin-25-le_petit_rien_xiveme_ordre_couperin-88-1
17.mp3.
Error opening input files: Invalid data found when processing input
 54%|█████▎    | 22/41 [00:14<00:12,  1.48it/s, loss=2.4]   [mp3 @ 0x79f414000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x79e824000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/6/norine_braun-now_and_zen-08-gently-117-146.mp3.
Error opening input files: Invalid data found when processing input

epoch 6/8 | train 2.3690 | val 2.3814 | time 273.7s
 34%|███▎      | 122/364 [01:23<02:38,  1.52it/s, loss=2.37][mp3 @ 0xafac3c000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xafb038000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/9/american_baroque-dances_and_suites_of_rameau_and_couperin-25-le_petit_rien_xiveme_ordre_couperin-88-1
17.mp3.
Error opening input files: Invalid data found when processing input
 50%|████▉     | 181/364 [02:03<01:59,  1.53it/s, loss=2.37][mp3 @ 0xaf3020000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xaf2c58000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/8/jacob_heringman-josquin_des_prez_lute_settings-19-gintzler__pater_noster-204-233.mp3.
Error opening input files: Invalid data found when processing input
 88%|████████▊ | 322/364 [03:38<00:28,  1.46it/s, loss=2.37][mp3 @ 0xc04c60000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xc05014000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/e/solace-balance-04-djinn-146-175.mp3.
Error opening input files: Invalid data found when processing input
 54%|█████▎    | 22/41 [00:14<00:12,  1.54it/s, loss=2.4]   [mp3 @ 0x99101c000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x990c5c000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/6/norine_braun-now_and_zen-08-gently-117-146.mp3.
Error opening input files: Invalid data found when processing input

epoch 7/8 | train 2.3713 | val 2.3752 | time 272.8s
 10%|▉         | 36/364 [00:23<03:35,  1.52it/s, loss=2.34][mp3 @ 0x758c40000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x758c3c000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/8/jacob_heringman-josquin_des_prez_lute_settings-19-gintzler__pater_noster-204-233.mp3.
Error opening input files: Invalid data found when processing input
 32%|███▏      | 116/364 [01:15<02:36,  1.58it/s, loss=2.36][mp3 @ 0xb2cc64000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xb2cc60000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/e/solace-balance-04-djinn-146-175.mp3.
Error opening input files: Invalid data found when processing input
 51%|█████     | 184/364 [01:59<01:58,  1.52it/s, loss=2.36][mp3 @ 0xa74c64000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0xa74c60000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/9/american_baroque-dances_and_suites_of_rameau_and_couperin-25-le_petit_rien_xiveme_ordre_couperin-88-1
17.mp3.
Error opening input files: Invalid data found when processing input
 54%|█████▎    | 22/41 [00:14<00:12,  1.51it/s, loss=2.4]   [mp3 @ 0x9cac60000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x9cac5c000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/6/norine_braun-now_and_zen-08-gently-117-146.mp3.
Error opening input files: Invalid data found when processing input

epoch 8/8 | train 2.3561 | val 2.3740 | time 259.3s
```

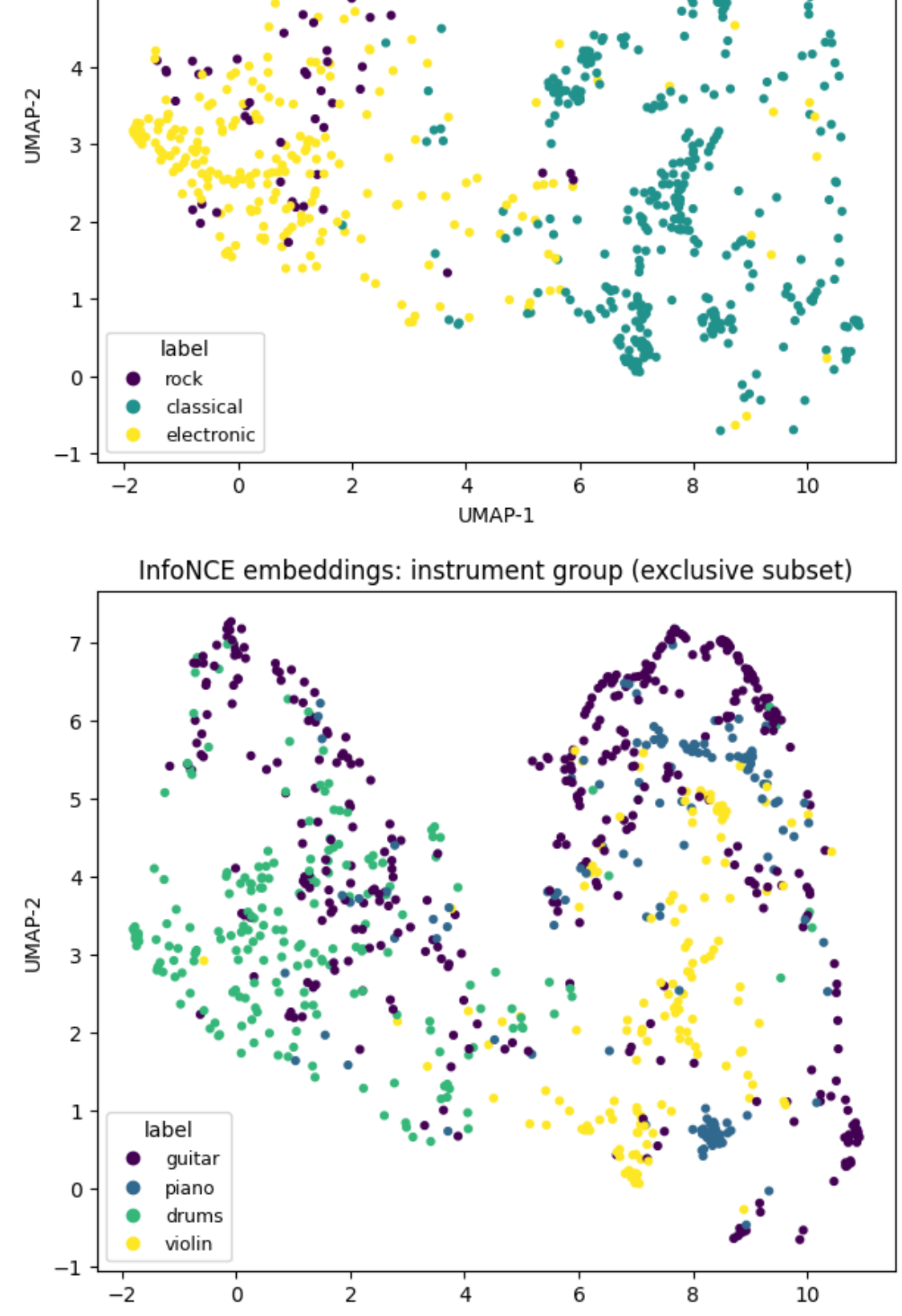
Step 4 — Embeddings, UMAP, and clustering (InfoNCE)

In [25]:

```python
Z_val_infonce, Y_val2, T_val2 = extract_track_embeddings(
    infonce_model,
    val_ds,
    chunks_per_track=2
)

reducer2 = umap.UMAP(n_components=2, random_state=SEED)
Z2_infonce = reducer2.fit_transform(Z_val_infonce)

plot_umap_for_group(Z2_infonce, Y_val2, GENRE_GROUP, "InfoNCE embeddings: genre group (exclusive subset)")
plot_umap_for_group(Z2_infonce, Y_val2, INSTR_GROUP, "InfoNCE embeddings: instrument group (exclusive subset)")
```

```
embedding:  57%|█████▋    | 1467/2586 [00:25<00:18, 58.97it/s][mp3 @ 0x9d4c68000] Failed to find two consecutive MPEG audio frames.
[in#0 @ 0x9d4c64000] Error opening input: Invalid data found when processing input
Error opening input file /Users/meco/MBP Files/Gits/ml-for-sound-and-music/datalabs/audio/6/norine_braun-now_and_zen-08-gently-117-146.mp3.
Error opening input files: Invalid data found when processing input
/Users/meco/MBP Files/Gits/upfs-virtualenv/lib/python3.11/site-packages/umap/umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting random_state. Use no seed for parallelism.
  warn(
```





Training diagnostics and environmental footprint

Summarized final:

- train/val losses per epoch
- total training time
- estimated CO2 emissions

- resource snapshots (CPU/RAM and GPU memory when available)

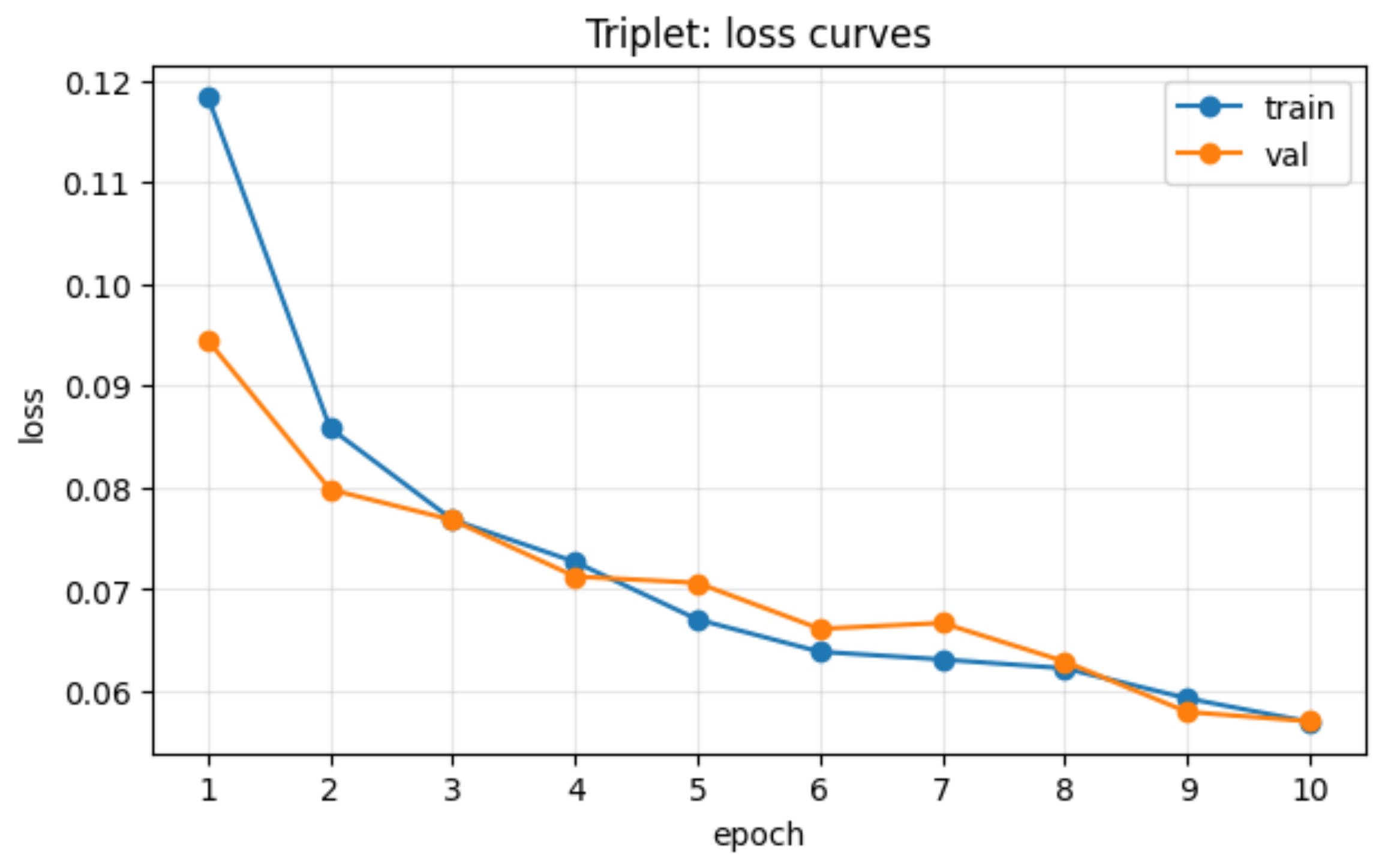
In [21]:

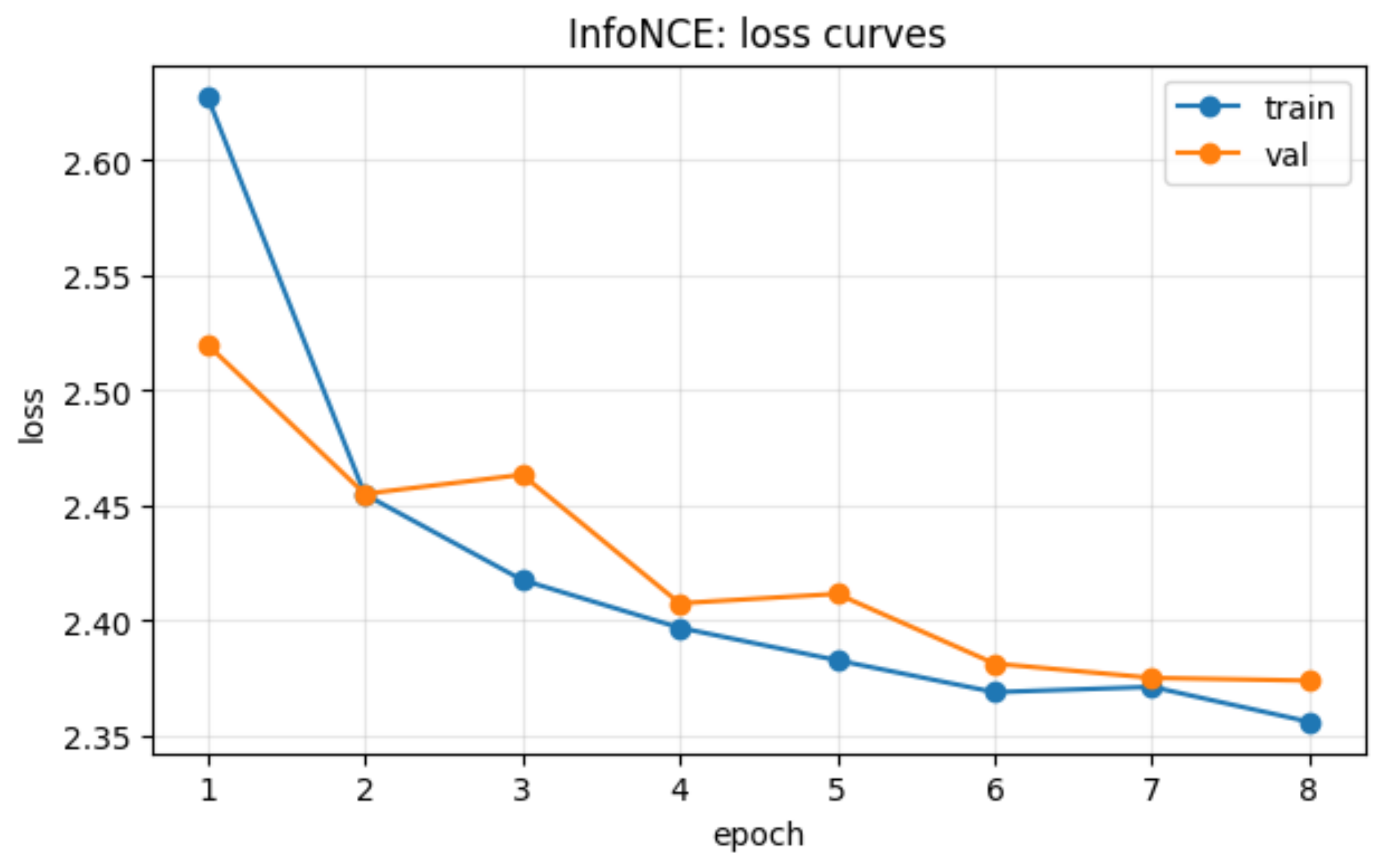
```python
print("triplet train_loss:", triplet_hist["train_loss"])
print("triplet val_loss:", triplet_hist["val_loss"])
print("infonce train_loss:", infonce_hist["train_loss"])
print("infonce val_loss:", infonce_hist["val_loss"])
```

```
triplet train_loss: [0.11843098943623212, 0.08585070981388236, 0.07684566922854263, 0.07271067061267056, 0.067061441070349, 0.06388972458624087, 0.06313129322562408, 0.06226352363272191, 0.0593
11812170422994, 0.05700704139135369]
triplet val_loss: [0.09448156433134544, 0.07982088034836257, 0.07682311371332262, 0.07127474820831926, 0.07065354996338123, 0.06614153250688459, 0.06671577410363569, 0.06293194631977779, 0.0579
7789900041208, 0.05706441702275741]
infonce train_loss: [2.6278739073774315, 2.4548961349896024, 2.4176964373378964, 2.3969472417464623, 2.3828384909001024, 2.368966560442369, 2.371262471099476, 2.356073702429677]
infonce val_loss: [2.520007784773664, 2.4551274660156994, 2.463430823349371, 2.407582294650194, 2.4116771977122236, 2.381399846658474, 2.375179337292183, 2.3739833279353815]
```

In [26]:

```python
import numpy as np
import matplotlib.pyplot as plt

def plot_history(hist, title):
    tr = hist.get("train_loss", [])
    va = hist.get("val_loss", [])
    n = min(len(tr), len(va))

    if n == 0:
        print("No hay datos para graficar en:", title)
        print("train_loss:", tr)
        print("val_loss:", va)
        return

    epochs = np.arange(1, n + 1)

    plt.figure(figsize=(7,4))
    plt.plot(epochs, tr[:n], marker="o", label="train")
    plt.plot(epochs, va[:n], marker="o", label="val")
    plt.title(title)
    plt.xlabel("epoch")
    plt.ylabel("loss")
    plt.xticks(epochs)
    plt.grid(True, alpha=0.3)
    plt.legend()
    plt.show()

%matplotlib inline
import matplotlib.pyplot as plt

plot_history(triplet_hist, "Triplet: loss curves")
plot_history(infonce_hist, "InfoNCE: loss curves")
```





Economic analysis (training and inference)

operational cost estimated from:

- energy consumption proxy (if available from CodeCarbon logs)
- electricity price (€/kWh)
- cloud compute pricing (optional reference scenario)

Reported:

- estimated energy cost
- rough equivalent of cloud cost for a typical GPU instance duration

In [23]:

```python
# economic analysis (simple)
ELECTRICITY_EUR_PER_KWH = 0.25

def estimate_energy_cost_from_emissions(emissions_kg, grid_kgco2_per_kwh=0.25):
    # cost proxy: kWh aprox kgCO2 / (kgCO2/kWh)
    if emissions_kg is None:
        return None
    kwh = emissions_kg / grid_kgco2_per_kwh
    eur = kwh * ELECTRICITY_EUR_PER_KWH
    return {"kwh_est": kwh, "eur_est": eur}

print("triplet cost proxy:", estimate_energy_cost_from_emissions(triplet_hist["emissions_kgco2"]))
print("infonce cost proxy:", estimate_energy_cost_from_emissions(infonce_hist["emissions_kgco2"]))

# cloud-equivalent rough estimate
GPU_HOURLY_EUR = 1.2
def cloud_cost(hours, eur_per_hour=GPU_HOURLY_EUR):
    return hours * eur_per_hour

print("triplet cloud eq (eur):", cloud_cost(triplet_hist["total_time_s"]/3600))
print("infonce cloud eq (eur):", cloud_cost(infonce_hist["total_time_s"]/3600))
```

```
triplet cost proxy: {'kwh_est': 0.015013952641687882, 'eur_est': 0.0037534881604219705}
infonce cost proxy: {'kwh_est': 0.01239620876493703, 'eur_est': 0.003099052191234257З}
triplet cloud eq (eur): 0.8620516495704651
infonce cloud eq (eur): 0.711772616068522
```

In [24]:

```python
# Colab: browser notification when finished
import sys

if "google.colab" in sys.modules:
    from google.colab import output
    output.enable_custom_widget_manager()
    output.eval_js("new Notification('Notebook finished', {body: 'Lab run completed.'});")
else:
    print("Not running on Colab.")


print("\n" + "="*60)
print("NOTEBOOK FINISHED. Lab done executing")
print("="*60 + "\n")

# Try beep
try:
    import IPython
    from IPython.display import Audio, display
    display(Audio([1,0]*2000, rate=8000, autoplay=True))
except Exception as e:
    print("Beep not available:", e)
```

```
Not running on Colab.

============================================================
NOTEBOOK FINISHED. Lab done executing
============================================================
```

Short report

Decisions

- dataset: MagnaTagATune (MTAT) via mirdata, 3-second random chunks at 16 kHz
- model: lightweight CNN encoder with L2-normalized embeddings (64-D)
- augmentations: low-cost waveform noise + gain to keep runtime low on CPU/MPS
- training: 10 epochs for triplet (hard negatives) and 5 epochs for InfoNCE (temp=0.2)

Observations

- triplet vs InfoNCE: compare separation in UMAP plots for genre and instrument groups
- clustering: k-means indicates whether embedding geometry aligns with semantic groupings
- multi-label challenge: exclusive subsets were used to build non-overlapping tag plots

Resource and sustainability

- runtime and emissions were tracked with CodeCarbon (offline)
- CPU/RAM snapshots were stored per epoch, GPU memory included when available

Limitations and next steps

- more epochs and stronger augmentations may improve invariances but increase cost
- larger batch sizes benefit InfoNCE but may be limited on MacBook
- a linear probe on tags could quantify embedding usefulness beyond visualization